GW170817 confines vestigial susceptibility to a finite interval under an assumed graviton speed law

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The conservative GW170817 cap $|\Delta c/c| \leq \tau = 3 \times 10^{-15}$ confines $\chi_{\rm vest} \geq 0$ to $[(1-\tau)^2, (1+\tau)^2]$ under the assumed leading-order law $c_{\rm GW} = c\sqrt{\chi_{\rm vest}}$ and source-emission assumptions. Here c is the electromagnetic reference speed; the free susceptibility $\chi_{\rm vest}$ models normalized dimensionless metric response. The interval has width 4τ, contains 1, and is over fourteen orders of magnitude narrower than adopted $[0.1, 10]$. Microscopic mode, normalization and range are not derived; only this kinematic model is constrained.

INTRODUCTION

Spinor-gravity constructions use composite vielbeins built from fermions [6, 8]. Volovik proposes a vestigial phase with metric order preceding a nonzero tetrad expectation [1]. His symmetry-breaking analysis distinguishes absorbed Nambu–Goldstone modes from the Higgs-sector gravitational-wave modes, whose counting also uses coordinate constraints [2]. Here we assume that a metric-channel excitation of such a substrate can be identified with a spin-2 second-sound graviton.

Motivation and assumed speed law. Volovik applies Landau two-fluid hydrodynamics to a de Sitter state in [3]. The underlying thermodynamic framework treats the gravitational normal component as stiff matter [4]. Within that de Sitter model, the Landau formula and the stiff-component thermodynamic relations give a second-sound speed $s_2 = c$, independently of the de Sitter Hubble parameter H. He discusses several possible graviton interpretations of this collective mode and leaves its type open. For the spin-2 substrate identification assumed here, we also assume the susceptibility-dependent speed law:

$$c_{\rm GW}^2(\chi_{\rm vest}) = c^2 \cdot \chi_{\rm vest}, \qquad c_{\rm GW}(\chi_{\rm vest}) = c\sqrt{\chi_{\rm vest}}, \quad (1)$$

where $\chi_{\rm vest}$ is the dimensionless, normalized metric-channel response of the assumed vestigial substrate. It is a free parameter in this calculation. The random phase approximation (RPA) resummation models it using unevaluated bare susceptibility and coupling inputs, so that form alone supplies no numerical value of $\chi_{\rm vest}$. The underlying lattice-fermion framework is the one whose unitarity Vergeles establishes [10]. Equation (1) is an assumed kinematic speed law. We constrain its free susceptibility without deriving a propagating second-sound mode from a microscopic action, its propagator, or its coupling to matter. This missing dynamical bridge is a limitation of the present calculation, not an impossibility result.

The 2017 binary-neutron-star merger GW170817 [11] yielded a tight propagation constraint under the collaboration's conservative distance and emission-time assumptions. Comparing the gravitational-wave arrival time at LIGO/Virgo with the gamma-ray signal at Fermi/INTEGRAL gives $-3 \times 10^{-15} \leq \Delta c/c \leq +7 \times 10^{-16}$ under those assumptions. We use its conservative symmetric envelope, $|\Delta c/c| \leq \tau$ with $\tau = 3 \times 10^{-15}$. This cap inherits the assumed source-emission window; alternative emission histories can weaken the inferred propagation constraint.

Our assumed speed law satisfies the GW170817 bound only on a sub-interval of our adopted range $\chi_{\rm vest} \in [0.1, 10]$ narrower than that range by more than 14 orders of magnitude. The GW170817-compatible window $[(1-\tau)^2, (1+\tau)^2]$ has width 4τ in $\chi_{\rm vest}$ and sits strictly interior to our adopted range $[0.1, 10]$ about the anchor $\chi_{\rm vest} = 1$. Its relative width is $4\tau/9.9 \approx 1.2 \times 10^{-15}$; a probability mass additionally requires a specified prior. The result is a fine-tuning constraint on one specific kinematic identification, not a refutation of emergent gravity nor of the Akama–Diakonov–Wetterich (ADW) framework. Three possible mechanisms are discussed below.

Relation to the 2017 constraint sweep. Using GW170817 to eliminate a gravity model is not new. Within months of the detection, four Letters analyzed constraints on specified gravitational actions and cosmological models. Baker *et al.* [12] constrained scalar-tensor and vector-tensor theories and obtained a conditional graviton-mass bound in bimetric gravity. Creminelli and Vernizzi [13] required luminal propagation to remain robust on nearby cosmological backgrounds to avoid tuning, reducing the three independent quartic and quintic beyond-Horndeski functions to one. Sakstein and Jain [14] combined the GW bound with cluster and stellar tests to constrain beyond-Horndeski parameters; their separate combination of GW and M87 black-hole constraints made the covariant quartic galileon cosmologically irrelevant for the order-unity matter couplings considered. Ezquiaga and Zumalacárregui [15] also discussed constraints on GW-speed-modifying couplings in Einstein-aether, some sectors of Hořava gravity, and Generalized Proca theories, and the impact on TeVeS and MOND-like theories.

These analyses constrain specified actions, EFT oper-

ators and cosmological models under assumptions about backgrounds and matter coupling. This Letter instead constrains the susceptibility in the assumed $c_{\rm GW} = c\sqrt{\chi_{\rm vest}}$ identification. It supplies no microscopic Lagrangian for that identification and constrains no microscopic operator coefficients. Its roughly one-part-in-10^{15} width comparison within the adopted order-unity window is neither a prediction of $\chi_{\rm vest}$ nor a prior-independent probability.

A microscopic realization must determine whether its operator-level mechanisms preserve or alter the assumed speed law and its applicability. The Discussion considers possible mechanisms; the finite-tolerance constraint alone derives none of them.

FORMAL SETUP AND RESULT

Adopted susceptibility comparison window. Our substrate models the vestigial-phase metric-channel susceptibility by the RPA *resummation* $\chi_{\rm RPA} = \chi_0/(1-\gamma_\star\chi_0)$ in the standard Hertz–Millis convention. Here χ_0 is the bubble integral; $\chi_{\rm RPA}$ is the resummed response and $\gamma_\star$ the effective coupling; physical normalization remains undetermined. The formalized closed form establishes stability properties but evaluates neither χ_0 nor $\gamma_\star$, so it fixes no numerical $\chi_{\rm vest}$, which is why $\chi_{\rm vest}$ enters Eq. (1) as a free parameter. We adopt the comparison window $\chi_{\rm vest} \in [0.1, 10]$, centered geometrically on the chosen exactly luminal anchor $\chi_{\rm vest} = 1$. Neither the RPA identity nor dimensional analysis determines the physical normalization, center or endpoints. Constructing the physical metric-channel response, including its normalization and domain, and establishing the susceptibility window from the microscopic theory remain required. We emphasize the provenance: the window is a modeling choice, not a published derivation: the Vergeles lattice analysis [10] establishes lattice unitarity for degree-of-freedom-preserving evolution under its stated function-space assumptions. It does not compute $\chi_{\rm vest}$ or assign it a range. The comparison window also occurs in the companion study of the linearized Einstein equations [16] for $\alpha_{\rm ADW}$, its free dimensionless coefficient multiplying an adopted induced-Newton baseline. We use that study's $\alpha_{\rm ADW} \in [0.1, 10]$ range as an order-of-magnitude comparison, without identifying $\alpha_{\rm ADW}$ with $\chi_{\rm vest}$ or deriving either range.

The leading-order $c_{\rm GW}$. Under the assumed law (1), a positive susceptibility and physical reference speed $c > 0$ give $c_{\rm GW} > 0$. For every $c \neq 0$, the relative speed deviation is

$$\Delta c/c(\chi_{\rm vest}) \equiv \frac{c_{\rm GW}(\chi_{\rm vest}) - c}{c} = \sqrt{\chi_{\rm vest}} - 1.$$

On the nonnegative domain this deviation is strictly increasing and vanishes if and only if $\chi_{\rm vest} = 1$. Thus unity is the exactly luminal anchor; finite-tolerance compatibility requires an interval rather than this single value.

Finite-tolerance compatibility. Fix $\tau = 3 \times 10^{-15}$ for the conservative symmetric cap.

Theorem 1 (Finite-tolerance compatibility)**.** *For $\chi_{\rm vest} \geq 0$, $|\Delta c/c(\chi_{\rm vest})| \leq \tau \iff \chi_{\rm vest} \in [(1-\tau)^2, (1+\tau)^2]$.*

Indeed, the cap is equivalent to $1-\tau \leq \sqrt{\chi_{\rm vest}} \leq 1+\tau$. Since $0 < \tau < 1$, both bounds are nonnegative, so squaring gives the stated interval. Conversely, monotonicity of the square root recovers the cap from these squared bounds.

Every compatible susceptibility therefore satisfies

$$0.1 < (1-\tau)^2 \leq \chi_{\rm vest} \leq (1+\tau)^2 < 10.$$

The compatible interval contains $\chi_{\rm vest} = 1$ and has nonzero width exactly $(1+\tau)^2 - (1-\tau)^2 = 4\tau = 1.2 \times 10^{-14}$. The adopted range has width 9.9, and the strict comparison is

$$10^{14}\left[(1+\tau)^2 - (1-\tau)^2\right] < 9.9.$$

This is a comparison of interval widths within an adopted domain. It supplies no probability without a prior: under a uniform prior on $[0.1, 10]$, the compatible interval has probability mass $4\tau/9.9$.

Endpoint checks. The endpoints provide quantitative checks of the confinement.

Theorem 2 (Lower endpoint)**.** *At $\chi_{\rm vest} = 0.1$, $\Delta c/c = \sqrt{0.1} - 1 \approx -0.6838$, hence $|\Delta c/c|/\tau \approx 2.28 \times 10^{14}$ and the conservative symmetric cap is violated.*

Theorem 3 (Upper endpoint)**.** *At $\chi_{\rm vest} = 10$, $\Delta c/c = \sqrt{10} - 1 \approx +2.1623$, hence $|\Delta c/c|/\tau \approx 7.21 \times 10^{14}$ and the conservative symmetric cap is violated.*

The exclusions already follow from $\sqrt{1/10} \leq \sqrt{1/4} = 1/2$ and $\sqrt{10} \geq \sqrt{4} = 2$. Both endpoints therefore fail simultaneously, while the finite interval in Theorem 1 remains compatible. Figure 1 shows this distinction.

Reference-speed independence of the endpoint constraints. The endpoint exclusion also applies directly to the relative deviation of the speed itself:

$$\left|\frac{c_{\rm GW}(\chi_{\rm vest}) - c}{c}\right| > \tau, \qquad \chi_{\rm vest} \in \{0.1, 10\}, \quad c \neq 0.$$

This includes the physical domain $c > 0$ and follows from the relative-deviation identity above. Changing c therefore cannot make either endpoint compatible under the assumed law.

Physical scope. This two-point exclusion leaves the compatible interior interval intact. Neither it nor the c-free endpoint statement quantifies over microscopic actions, propagators or matter couplings, so neither establishes the absence or impossibility of a dynamical derivation. A physical application still requires the response

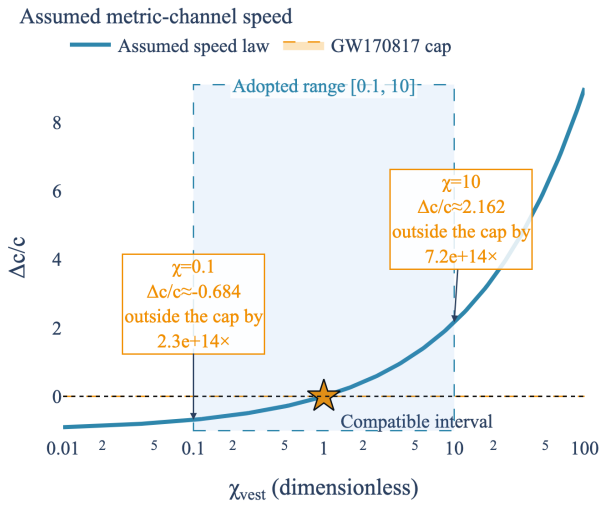


FIG. 1. **Headline result.** For the assumed speed law, the $\Delta c/c = \sqrt{\chi_{\rm vest}} - 1$ deviation across the adopted comparison window $\chi_{\rm vest} \in [0.1, 10]$ (pale blue band, dashed blue border) versus the conservative symmetric envelope of the GW170817 [11] multi-messenger constraint, $|\Delta c/c| \leq 3 \times 10^{-15}$ (amber, drawn at $\Delta c/c = 0$ because the cap's own width is far below one pixel at this scale). Over this comparison window $\Delta c/c$ sweeps $[-0.68, +2.16]$, exceeding the cap by $\sim 10^{14}\times$ *at the endpoints* (orange callouts) and vanishing at $\chi_{\rm vest} = 1$. The compatible $\chi_{\rm vest}$ window $[(1-\tau)^2, (1+\tau)^2]$ has width $4\tau = 1.2 \times 10^{-14}$, compared with the adopted range's width 9.9. Its finite extent is unresolved at this scale; the gold star marks its anchor $\chi_{\rm vest} = 1$, not a singleton compatible set. The window sits *inside* the band: the datum confines $\chi_{\rm vest}$, it does not exclude the range.

normalization and domain, the physical state, a propagating tensor mode relative to the same matter/light reference, and controlled transport and frequency matching with units and error bounds.

DISCUSSION

The conservative symmetric cap confines the assumed leading-order identification to $\chi_{\rm vest} \in [(1-\tau)^2, (1+\tau)^2]$. Its endpoints $1 \pm 2\tau + \tau^2$ allow susceptibility offsets of approximately $\pm 6 \times 10^{-15}$ for a speed-deviation tolerance of $\pm 3 \times 10^{-15}$. Exact luminality anchors this finite interval at $\chi_{\rm vest} = 1$; the measurement does not require an exact fixed point or exclude generic ultraviolet theories.

A separate Kerr–Schild construction uses prescribed null directions for analog backgrounds (`admitsKerrSchildForm_all`). The single-copy covector remains null for every scalar profile (`isVortexLikeChargeDistribution_all`). The polariton predicate combines these algebraic properties (`isConnectedToPolaritonRingdown_polariton`). Relating this construction to the speed test requires a dynamical ringdown model and a common matter and light reference.

Three possible mechanisms illustrate the unresolved microscopic choices. First, a deep-infrared symmetry, such as Lorentz invariance of the emergent metric sector, could make unity an exact fixed point of the renormalization-group flow. Such a mechanism would place $\chi_{\rm vest}$ inside the compatible interval through symmetry; whether it occurs in the ADW broken phase remains unproved. Second, a distinct ultraviolet-input metric channel could replace the second-sound identification and have dynamics that pin its susceptibility to unity. This is a model-building proposal to be investigated through the nonlinear emergent action. The missing dynamical bridge supplies no reason to prefer it. The finite-tolerance result derives neither mechanism.

Third, frequency dependence or a compensating correction in the LIGO band could modify the assumed frequency-independent law (1). The product ansatz $\delta_{\rm disp} = \gamma\omega$ is algebraic, not a derived microscopic derivative expansion. Its formal probe $\omega = 100$ requires a frequency convention and unit normalization before it can represent 100 Hz. The proposed $\gamma = \Gamma_H / c_{\rm GW}^2$ is unmatched: the same-symbol horizon damping rate in the acoustic setting establishes no physical identification with this metric-channel parameter. The chosen placeholder $|\gamma| \leq 10^{-30}$ bounds the correction below the cap at that numerical probe; it is neither measured nor derived, and the physical correction could be larger. This bounds only $\delta_{\rm disp}$, not the total deviation. In an additive model, an oppositely signed correction can restore compatibility for offsets just outside the cap. The full additive exclusion criterion, near-boundary restoration argument and supporting checks are given in the appendix. Deriving the dissipative parameter and the normalization from microscopic dynamics remains open.

Comparison with the induced Newton coupling. The in-preparation companion study [16] uses the emergent Newton-coupling parameterization $G_{\rm N,emerg} = \alpha_{\rm ADW}\, 12\pi/(N_f \Lambda_{\rm UV}^2)$, where N_f is a fermion-species-count parameter. In units $\hbar = c = 1$, the observed unreduced Planck mass is $M_{\rm Pl} = (G_{\rm N,obs})^{-1/2}$, with $G_{\rm N,obs}$ the observed Newton constant. At the adopted cutoff $\Lambda_{\rm UV} = M_{\rm Pl}$, the exact algebraic match condition is $\alpha_{\rm ADW}^\star = N_f/(12\pi)$. For the companion's stated count choices, the reported rounded matching values span $[0.40, 1.27]$, inside its adopted $[0.1, 10]$ range. Its kernel's Dirac-count convention and these Weyl-count choices require reconciliation before this comparison supplies a physical Standard Model normalization.

The companion proves the matching locus and properties of a candidate coefficient function; it does not extract the microscopic coefficient that selects the locus. That coefficient and $\chi_{\rm vest}$ describe distinct response normalizations. Our adopted interval contains unity but derives no value of $\chi_{\rm vest}$. The GW170817 cap confines this param-

eter to the much narrower compatible interval without supplying a microscopic selection mechanism.

Physical implication. Under the assumed speed law and source-emission conditions, GW170817 selects a narrow finite interval of normalized susceptibility, not a singleton. The adopted comparison range is no derived physical prior, and its width ratio supplies no probability without a prior. A physical realization still requires the actual response normalization and domain, the physical state, and a propagating tensor mode relative to the same matter/light reference, together with controlled frequency and transport matching, units and error bounds. The interval constrains the stated mathematical model while leaving its microscopic realization open.

Supporting formal results and reproducibility

This appendix gives the auxiliary predicate and frequency-correction checks used to delimit the Letter's conditional interval result. It preserves their domains and counterexamples; none supplies the missing microscopic mode, response normalization or physical matching. At the fixed $\tau = 3 \times 10^{-15}$, write $\texttt{LigoSatisfied}(\delta) \equiv |\delta| \leq \tau$.

Predicate audit and counterexamples

The proposed identification imposes positivity (P1), compatibility with the cap (P2), and sub-half-deviation (P3'):

$$\begin{aligned}&\texttt{H_VestigialModeIsGraviton}(\chi_{\mathrm{vest}})\\ &\quad\equiv \chi_{\mathrm{vest}} > 0 \ \wedge\ \texttt{LigoSatisfied}(\Delta c/c(\chi_{\mathrm{vest}}))\\ &\qquad\wedge\ |\Delta c/c(\chi_{\mathrm{vest}})| < 1/2.\end{aligned}$$

These three clauses are not independent. At the fixed tolerance, P2 implies P3' because $\tau < 1/2$. On the nonnegative domain, Theorem 1 also places every P2-compatible susceptibility in a strictly positive interval, so P1 holds there. Consequently, the complete compatible set for these conditions is precisely that same finite interval, including the anchor $\chi_{\mathrm{vest}} = 1$.

Four counterexamples distinguish compatibility from positivity alone. The endpoints 0.1 and 10 refute the conjunction through failure of P2. At zero, P1 fails; this point lies outside the adopted range. At the interior point $\chi_{\mathrm{vest}} = 1/4$, $\Delta c/c = -1/2$: P1 holds, but P2 and the strict P3' inequality both fail. The quarter-point example therefore shows that P1 alone does not imply P3'. These counterexamples do not establish independence of the three clauses or isolate a uniquely failing clause. The corresponding definitions and algebraic results are recorded in `GravitationalWaves.lean`, Sections 1–4.

Illustrative frequency correction and additive bound

Equation (1) is an assumed frequency-independent kinematic law. A frequency-dependent susceptibility or a compensating correction in the LIGO band is a possible phenomenological extension. We consider the algebraic ansatz $\delta_{\mathrm{disp}} = \gamma\omega$, without asserting that it is the leading term of a matched microscopic derivative expansion. The formal probe is the real number $\omega = 100$. Interpreting it as a 100 Hz band requires a physical frequency convention and unit normalization. Identifying $\gamma = \Gamma_H/c_{\mathrm{GW}}^2$ introduces a proposed, unmatched dissipative parameter Γ_H in this metric channel. The horizon damping rate denoted by the same symbol in the acoustic setting does not establish that physical identification. Under the assumed coefficient bound $|\gamma| \leq 10^{-30}$, $|\delta_{\mathrm{disp}}|$ at this numerical probe is below the GW170817 tolerance. The adopted 10^{-30} is a placeholder chosen to keep the correction below the cap; it is neither derived nor measured. In the further additive model $\delta_{\mathrm{tot}} = \delta_0 + \delta_{\mathrm{disp}}$, with $\delta_0 = \sqrt{\chi_{\mathrm{vest}}} - 1$, that assumed coefficient bound gives $|\delta_{\mathrm{disp}}| \leq B = 10^{-28}$ at $\omega = 100$. The reverse triangle inequality then excludes restoration of $|\delta_{\mathrm{tot}}| \leq \tau$ whenever $|\delta_0| > \tau + B$. Offsets just outside $\pm\tau$ can nevertheless become compatible through an oppositely signed correction within this bound. Larger corrections are not excluded by the placeholder theorem. Deriving Γ_H and the normalization of this ansatz from microscopic dynamics remains open.

For a positive numerical probe ω, `dispersion_within_ligo_iff` gives $|\gamma\omega| \leq \tau \iff |\gamma| \leq \tau/\omega$ at the fixed GW170817 tolerance τ, bounding the correction alone. `dispersion_correction_abs_bound`, `dispersion_correction_linear_in_gamma` and `dispersion_correction_zero_at_no_dissipation` describe the product's magnitude, additivity in γ and zero-coefficient limit.

`vestigial_dispersion_below_ligo_at_inspiral_peak` records the conditional bound at the numerical probe used above.

Theorem provenance and recorded validation

The finite-interval theorem is `c_GW_match_iff_chi_close_to_one`; the lower and upper endpoint results are `natural_lower_violates_ligo` and `natural_upper_violates_ligo`. The endpoint statement for every real $c \neq 0$ is `natural_endpoints_violate_ligo_for_every_c`. The RPA formalization discussed in the Letter is `VestigialSusceptibility.lean`; its unevaluated inputs remain necessary for physical normalization.

The following software and proof records support reproducibility of the stated algebraic calculation. They establish neither the microscopic response normalization nor a physical mode or state.

The result is reproducible in two layers. The

Lean layer is `GravitationalWaves.lean` (32 theorems, zero `sorry`, zero new axioms, zero `maxHeartbeats` overrides) at the project's pinned Mathlib4 commit; the Python layer is `src/gravitational_waves/` with 89 `pytest` cases cross-checking every numeric quoted in the abstract and figure caption. The headline figure (`src/core/visualizations.py:fig_c_GW_vs_ligo_constraint`) and its underlying numerics are committed-pinned.

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